

4. Research and Teaching Setup

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The Setup Tax

The old way

- Install the language, then the packages, then the editor
- Fight PATH variables, versions, and dependency errors
- Hours lost before a single line of research gets done

The new way

- Open Claude Code and say what you want to work in
- It installs the software, wires it up, and fixes what breaks
- You describe the environment; it builds it

This session is the research and teaching stack — Python, R, Stata, LaTeX, Quarto, Git, GitHub — and the single idea that Claude Code can stand all of it up for you.

VS Code: The Hub

One Editor for Everything

What it is

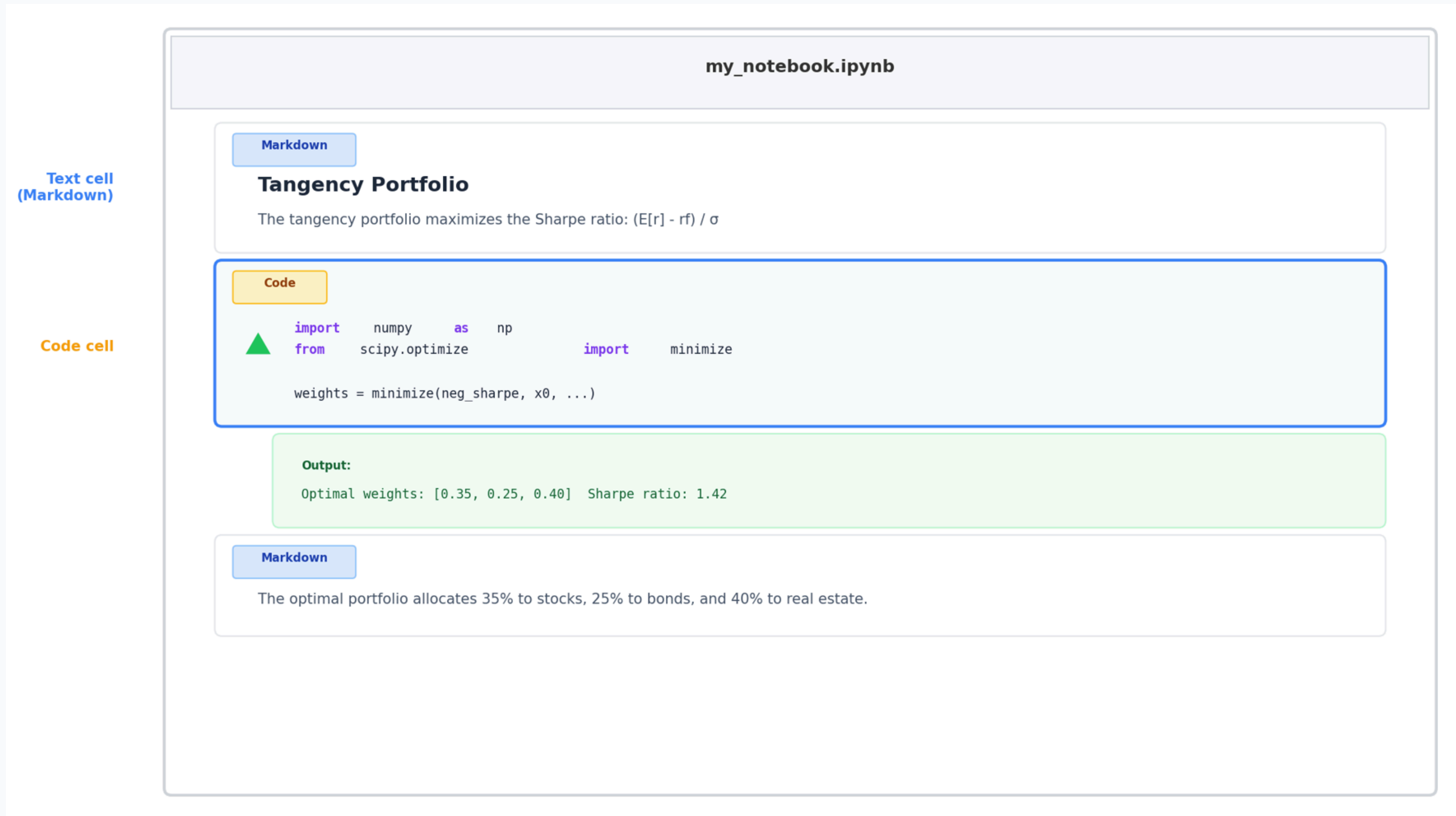
- A free editor that handles every language you use
- Python, R, Stata, LaTeX, Markdown, Quarto — all in one window
- Extensions add support for each; Claude installs the ones you need

Why it is the hub

- Claude lives in a side panel, aware of your open files
- Run code, edit papers, preview slides, commit to Git — without leaving
- Proposed edits appear as a **diff** you approve

Pick one home for your work. VS Code plus Claude is that home — everything below plugs into it.

Notebooks, Right in the Editor



A Jupyter notebook interleaves code, output, and prose — ideal for analysis you want to narrate. VS Code runs

Languages

Python and R

Python

- The default for data work, machine learning, and apps
- Claude installs libraries for you — `pip install pandas`
- On a Mac the command is `python3`; Claude adjusts automatically

R

- The statistician's language — and much of empirical social science
- Claude writes and runs `.R` scripts, and knows the tidyverse and common packages
- Same loop: describe the analysis, it writes the code and runs it

You do not have to choose a language on ideological grounds. Claude is fluent in both — work in what your field and your co-authors use.

Stata

Claude writes your do-files

- Describe the regression, the panel setup, the table — Claude writes the `.do` file
- It knows Stata syntax, `esttab`, `reghdfe`, and the usual toolkit
- Run it interactively, or let Claude run it in batch and read the log

The workflow

- Keep your data and do-files in a project folder
- “Add clustered standard errors and re-run” — it edits the do-file and executes
- The log becomes something Claude reads back and explains

Stata’s GUI is not the point of contact. The `.do` file is — and that is exactly the kind of text Claude writes, edits, and reruns well.

Writing

LaTeX

For papers that look right

- The standard for math-heavy papers, journals, and job-market work
- Claude drafts sections, builds tables, and fixes the errors LaTeX is famous for
- It reads the cryptic compile errors and resolves them — the worst part, gone

Tables from results

- Point it at your regression output; it produces the formatted LaTeX table
- Change a spec, regenerate the table — no hand-editing columns
- Figures, citations, and BibTeX handled the same way

The tedium of LaTeX — the debugging, the table formatting, the reference wrangling — is precisely what AI removes.

Quarto: One Source, Many Outputs

Write once

Plain-text `.qmd` files:
prose, code, and math
together

Render anywhere

The same source
becomes a paper
(PDF), a website, or
slides

These slides

This deck is Quarto —
rendered to reveal.js,
exportable to PDF or
PowerPoint

Quarto is the modern successor to R Markdown and works with Python, R, and Stata alike. Because it is plain text, Claude authors and restyles whole documents — as it did for every deck in this course.

Git & GitHub

Git: Track Changes for a Whole Project

The idea

- Git is **Track Changes** for an entire folder of files
- Every save you choose is a **commit** — a snapshot you can return to
- Break something? Roll back to any past version

With Claude, it is invisible

- Claude Code installs Git and commits for you
- "Commit this," "show me what changed," "go back to yesterday's version"
- The safety net that lets you say yes to bigger changes

The reason you can let AI edit your project freely is that Git remembers every state — nothing is ever truly lost.

GitHub: Backup, Sharing, Publishing

Your code in the cloud

- GitHub hosts your Git projects — backed up and shareable
- “Push this project to a new GitHub repo” — Claude sets up the CLI and does it
- Co-authors clone it; you track issues and versions together

GitHub Pages: free hosting

- Name a page `index.html`, push, flip on Pages
- Live at a `github.io` address — no server, no cost
- Perfect for a course site, a paper’s companion page, or a small app

For solo research, GitHub is mainly backup plus a launch pad. For teaching, it is free hosting for anything you want students to reach.

The One-Line Setup

What you tell Claude Code

```
Set up a research project in this folder. Install Python with pandas, numpy, statsmodels, and matplotlib; set up R; install Quarto and LaTeX. Create a project structure with folders for data, code, and output. Initialize Git and push it to a new private GitHub repo.
```

One paragraph. Claude installs each piece, verifies it, creates the structure, and hands you a working, version-controlled project — the setup tax, paid once, by the machine.

The Point

The environment is no longer something you assemble by hand and pray works. You **describe the stack you want** — languages, writing tools, version control — and Claude Code builds it, fixes it, and keeps it working. Your time goes to the research, not the setup.

Breakout

Questions to Discuss

- What is in your current research stack — and what part of setting it up costs you the most time?
- Which language do you and your co-authors actually work in? Where would AI help across that boundary?
- Do you use version control now? What has “losing work” cost you that Git would have prevented?
- What would you want on a free GitHub Pages site for one of your courses?